

CSA1304-THEORY OF COMPUTATION LAB EXPERIMENTS

3. Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$S \rightarrow 0A1$ $A \rightarrow 0A \mid 1A \mid \varepsilon$

AIM

Write a C program to check whether a given string belongs to the Context-Free Grammar (CFG)

CODE

```
#include <stdio.h>
#include <string.h>

int main()
{
    char str[100];
    int len, i, valid = 1;

    printf("Enter the binary string: ");
    scanf("%s", str);

    len = strlen(str);

    if(len < 2)
        valid = 0;

    for(i = 0; i < len; i++)
    {
        if(str[i] != '0' && str[i] != '1')
        {
            valid = 0;
            break;
        }
    }

    if(valid && str[0] == '0' && str[len-1] == '1')
        printf("String Accepted\n");
    else
        printf("String Rejected\n");
}
```

```
    return 0;  
}
```

Output

Clear

```
Enter the binary string: 01  
String Accepted
```

```
=== Code Execution Successful ===
```